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A method for detecting safety helmets underground based on the YOLOv11-SRA model

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Monitoring the safety of underground work environments is a fundamental aspect of industrial safety production. Detecting safety helmet wearing, as a critical task, poses numerous challenges, including interference caused by complex backgrounds, low-light conditions, and insufficient detection accuracy achieved for small targets. Existing methods exhibit limitations in terms of multiscale feature fusion, foreground localization accuracy, and dynamic context modeling. Traditional feature pyramids struggle to resolve cross-scale feature conflicts, whereas fixed dilated convolutions lack adaptability in complex scenes. To address these issues, a YOLOv11-SRA model based on the YOLOv11 architecture is proposed in this paper; the model is integrated with a three-stage optimization strategy: the SAConv module dynamically adjusts the dilation rates to capture multiscale contextual information, thereby enhancing the robustness of small target detection. The RCM employs rectangular self-calibrated attention to refine the foreground region, improving the boundary localization capabilities of the model. The ASFF module fuses multiscale features through adaptive spatial weighting to alleviate feature conflicts. The effectiveness of the algorithm was validated based on a publicly available underground safety helmet dataset, i.e., CUMT-HelmeT, where it achieved a mean average precision at an IoU of 0.50 (mAP50) of 84.2% and a recall of 79.9%, significantly outperforming the mainstream models.

Keywords Safety helmet detection, YOLOv11, SAConv module, RCM module, ASFF module

Underground safety helmet detection is a critical technological requirement for ensuring the safety of workers in underground environments such as mining and tunnel construction areas. Its significance is manifested in various dimensions, including accident prevention, regulatory compliance oversight, and intelligent mining practice advancement. As global industrialization accelerates and the scale of underground resource development expands, the harsh and complex working conditions experienced in coal mines, which are characterized by methane explosions, roof collapses, and mechanical injuries, pose substantial threats to worker safety[1]. Safety helmets, as the most fundamental head protection equipment for underground workers, effectively mitigate the risk of injuries caused by falling objects or collisions, serving as the primary line of defense for miners' safety. However, the challenging underground working environments, coupled with high work intensity levels and management difficulties, often results in workers having insufficient safety awareness, leading to instances of noncompliance with safety helmet regulations. The traditional manual inspection methods are not only inefficient but also susceptible to blind spots and human error, hindering their ability to provide comprehensive and continuous safety supervision. Therefore, the establishment of an intelligent and automated underground safety helmet detection system is essential for achieving real-time monitoring and early warnings of the safety protection statuses of workers, thereby fundamentally reducing the likelihood of safety incidents, ensuring miners' safety, and fostering the overall enhancement of safety production standards within the coal industry.

The traditional underground safety inspection methods primarily include manual inspections, sensor monitoring, and mechanical detection. Manual inspections depend on safety officers regularly entering underground work areas to verify the usage of safety equipment by workers. However, this method is hindered by its low inspection frequency, limited coverage, and significant subjectivity, and the safety officers themselves also face risks. Sensor monitoring techniques involve embedding RFID tags, accelerometers, or pressure sensors into safety helmets to track their wearing statuses. Although this approach offers some automated monitoring capabilities, it is plagued by high hardware costs, maintenance challenges, and susceptibility to environmental interference, rendering it ineffective in reliably identifying proper helmet-wearing practices[2]. Furthermore, the existing helmets require modifications, leading to elevated implementation costs and technical complexity, which complicates the task of scaling sensor-based methods for use in large-scale underground operations. Mechanical

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detection methods rely on access control systems or detection checkpoints, necessitating workers to undergo safety helmet inspections when passing through. This method can be executed only at specific locations and fails to provide continuous monitoring throughout the work process. In summary, these traditional methods are generally characterized by limitations such as poor real-time performance, low accuracy, high deployment costs, and weak adaptability, rendering them inadequate for addressing the refined, intelligent, and real-time safety management requirements of modern mines. Therefore, more advanced and efficient detection technologies are needed. In recent years, deep learning technology has made significant breakthroughs in the field of computer vision, and object detection algorithms based on convolutional neural networks have provided novel technical approaches for detecting underground safety helmets. Compared with the traditional methods, deep learning-based approaches offer substantial advantages, including robust feature extraction capabilities, excellent adaptability, and high detection accuracy. These methods can automatically learn complex feature patterns in images, thus effectively addressing the complexity and variability of underground environments.

Currently, the mainstream object detection algorithms are primarily classified into two categories: two-stage detectors (e.g., the R-CNN series[3] and single-stage detectors (e.g., the YOLO series, SSD[4], etc.). Two-stage detectors perform object detection through two steps: candidate region generation and precise classification. While they offer high detection accuracy, they also entail greater computational complexity. In contrast, single-stage detectors directly predict object locations and categories from feature maps, ensuring high detection accuracy while providing faster processing speeds, making them more suitable for real-time detection applications. In the realm of safety helmet detection, researchers have proposed various solutions based on different deep learning architectures: CenterNet-based approaches[5] significantly enhance the speed of tracking by constructing lightweight feature hierarchies; Faster R-CNN-based methods demonstrate strong performance in complex backgrounds but exhibit slower inference speeds[6]; methods based on SSDs achieve a favorable balance between speed and accuracy[7]; and YOLO-based methods have garnered considerable attention because of their exceptional real-time performance, with ongoing detection accuracy, processing speed, and model weight reduction improvements[8,9]. These deep learning-based methods, which are trained on extensive labeled datasets, can effectively identify safety helmet targets from various angles and under different lighting conditions. This ability provides robust technical support for achieving underground safety supervision and drives the advancement of intelligent safety detection technology in the mining field.

To enhance the safety management capabilities of underground workers, a YOLOv11-SRA network that is specifically designed to address the dynamic characteristics of complex underground environments is proposed in this paper. Building upon the YOLOv11 architecture, the model integrates adaptive feature fusion and dynamic convolution techniques to optimize the helmet detection process in underground settings, thereby significantly improving the performance of the original network in complex mining environments. The algorithm includes the following innovations: (1) the introduction of an SAConv (switchable dilated convolution) to replace the original convolution layer, which effectively extracts multiscale features from underground scenes, thereby improving both the detection speed and accuracy of the model; (2) the utilization of an RCM (rectangular self-calibration module) to optimize the neck network, enhancing the ability of the model to recognize the wearing statuses of underground safety helmets; and (3) the adoption of an ASFF (adaptive spatial feature fusion) module to optimize the detection head, which enhances the ability of the model to fuse feature maps at different scales and improves the accuracy of small safety helmet detection in underground environments.

Related work

In recent years, breakthroughs in deep learning algorithms have positioned underground visual perception systems as a prominent research focus in the field of object detection, resulting in significant technological advancements. The YOLO series, which is recognized as a group of leading single-stage object detection algorithms, has been extensively applied and continuously refined for safety helmet detection tasks because of its outstanding real-time performance and detection accuracy. Related research has predominantly concentrated on optimizing network architectures, enhancing the feature extraction process, facilitating multiscale object detection, and designing lightweight models, thereby establishing a multidimensional, synergistically evolving technical framework.

Underground mining environments are characterized by high dust concentrations, foggy conditions, uneven lighting, and dynamic interference, which severely degrade the quality of surveillance images. Dust and fog scatter light, reducing image contrast levels and creating nonuniform blurred areas[10]. Additionally, low-light conditions exacerbate target edge blurring and noise interference issues. Strong light reflections from underground equipment operations can also create speckle noise, further complicating the accuracy of target detection. The occlusion and shadow effects caused by dynamic targets, such as moving equipment and personnel, increase the difficulty of detection. These challenges impose greater demands on the robustness of image preprocessing algorithms and detection models. To address the unique challenges posed by underground working environments, researchers have optimized and improved the YOLO algorithms from various perspectives. Dong et al.[11] proposed the MH-YOLO algorithm, which enhances the backbone network by applying a convolutional block attention mechanism (CBAM), replacing some downsampling convolutions with MaxPooling modules to mitigate the impact of sample category imbalances, and utilizing the ZoomCat and Scalseq (ZAS) feature extraction modules to improve the detection accuracy achieved for small and overlapping targets. Yang Jiaru et al.[12] approached the issues of small object detection and environmental noise interference in underground environments by designing the PBSS-YOLOv8 model. They introduced a small object detection layer (P2) and implemented the BiFPN framework to facilitate bidirectional information transmission. Furthermore, they incorporated the SPD-Conv module to minimize the loss of small object information and integrated a lightweight attention mechanism (SGE) to alleviate environmental noise effects. Fu et al.[13] proposed the Yolov8n-FADS algorithm, which replaces the head section with attention sequence fusion (ASF) and introduces the P2 detection layer. They designed

the ASF-P2 structure to comprehensively extract image feature information and adopted the SEAM module to enhance the detection head, thereby improving its effectiveness in terms of addressing occlusion issues. Zhang et al.[14] approached the problem of blurred target images caused by heavy fog, dust, and low lighting conditions in underground mining environments by integrating the AOD-Net defogging network with an improved YOLOv5 detection method. This lightweight defogging preprocessing step significantly enhances the detection accuracy attained in complex environments.

To address the challenges posed by small object detection and multiscale feature extraction and fusion, numerous studies have focused on optimizing network architecture designs. Zhao et al.[15] proposed the BDC-YOLOv5 model, which enhances the small object detection process by adding a detection layer and introducing a bidirectional feature pyramid network (BiFPN) on top of YOLOv5 in combination with a convolutional block attention module (CBAM). This approach not only improves the detection ability of the model for small objects but also effectively reduces the false-positive and false-negative rates of the model. Fang et al.[16] developed an enhanced YOLOv5 model by integrating attention mechanisms at the end of the backbone and neck networks, replacing the original feature pyramid network with additional cross-layer edges featuring adaptive connection weights, and employing transfer learning for model training purposes, thereby achieving significant accuracy improvements and good generalization performance. Tan et al.[17] based their work on the YOLOv5 framework, adding functional detection scales to capture smaller targets and introducing DIoU-NMS to replace the traditional NMS method, thus providing improved detection accuracy while maintaining high real-time detection speeds. Wang et al.[18] built upon the YOLOv7-tiny framework, introducing an efficient multiscale attention (EMA) module in the backbone to capture detailed information, replacing the detection head with a dynamic Head (DyHead) featuring self-attention mechanisms, and adopting Wise-IoU with a dynamic nonmonotonic focus mechanism as the loss function, effectively enhancing the ability of the model to handle occlusion scenarios. Shao et al.[19] proposed the FM-YOLOv7 algorithm, which includes the design of an MBCA module to enhance the feature extraction capabilities of shallow networks. They constructed a multiscale feature fusion and efficient layer aggregation network to obtain semantic and detailed information at varying scales and adopted an efficient intersection over union as the bounding box regression loss function to accelerate the model convergence process. Song et al.[20] addressed the challenges posed by complex backgrounds and small object detection, which are exacerbated by distance differences in industrial environments, by designing an improved YOLOv8 network. Their approach introduced a dilationwise residual attention module, dilated spatial pyramid pooling, and a normalized Wasserstein distance loss function to enhance the detail capture and multiscale feature processing performance of the model. Lv et al.[21] proposed the YOLOv10-MSH model based on the latest YOLOv10 framework, which utilizes contrastive learning for data augmentation purposes and introduces a hybrid attention mechanism that integrates spatial and channel attention to improve its feature extraction ability, effectively meeting the requirements for detecting safety helmets in underground mining operations in real time.

Owing to the limited computational resources required in underground environments, lightweight detection algorithms have emerged as a critical research direction. Deng et al.[22] designed the mini lightweight YOLOv3 (ML-YOLOv3) method by integrating CSPNet and GhostNet to create a more efficient residual network, termed CSP-Ghost-ResNet. They also developed a lightweight multiscale feature extraction network, the PAN-CSP-Network, which exhibited significant advantages over YOLO5 in terms of detection performance. Du et al.[23] introduced the BLP-YOLOv10 model, optimizing the feature extraction procedure by adjusting the main channel parameters, implementing a sparse attention mechanism, and incorporating low-frequency enhancement filters. This approach markedly reduces the number of required parameters while preserving high detection accuracy, thereby enhancing its applicability in resource-constrained environments. Xing et al.[24] developed the GPD-YOLOv8 lightweight detection model, which incorporates a P2 detection layer into the YOLOv8 architecture to enrich semantic feature representation. Additionally, it integrates a lightweight Ghost module to replace the original backbone network and introduces a dynamic detection head (Dyhead) that uses an attention mechanism to extract key feature information. This strategy significantly enhances the resulting detection accuracy while maintaining a high processing speed and demonstrates robust performance under varying lighting conditions.

In summary, YOLO-based safety helmet detection methods have made significant progress in various areas. However, underground safety helmet detection faces several challenges. (1) Underground environments exhibit complex and variable lighting conditions, where low-light or highly reflective areas often lead to feature extraction failures; (2) safety helmets are represented as small targets in long-distance monitoring images and are frequently obstructed by equipment or personnel, which complicates the ability of the existing algorithms to reliably capture effective features; and (3) detection models must achieve low-power real-time operation on embedded devices, yet high-precision models often consume excessive computational resources, thereby limiting their practical applicability. To address these research gaps, an improved safety helmet detection model, YOLOv11-SRA, is explored in this paper to provide a more effective and practical solution for underground safety helmet detection.

Methods

Overall architecture

The YOLOv11-SRA model (illustrated in Fig. 1) preserves the fundamental architecture of YOLOv11, which is composed of four components: an input preprocessing mechanism, a backbone network, a neck network, and a detection head. The input section employs an adaptive anchor box-based dynamic generation algorithm to enhance the initial feature representations of low-light images and small targets, such as safety helmets. The backbone network integrates the RCM (rectangular self-calibration module) into the original CSPDarknet architecture, establishing a geometric feature extraction layer through the application of rectangular convolutional kernels arranged in rows and columns. The self-calibration mechanism spatially corrects feature distortions caused by the occlusion of underground equipment, while simultaneously replacing the standard convolution

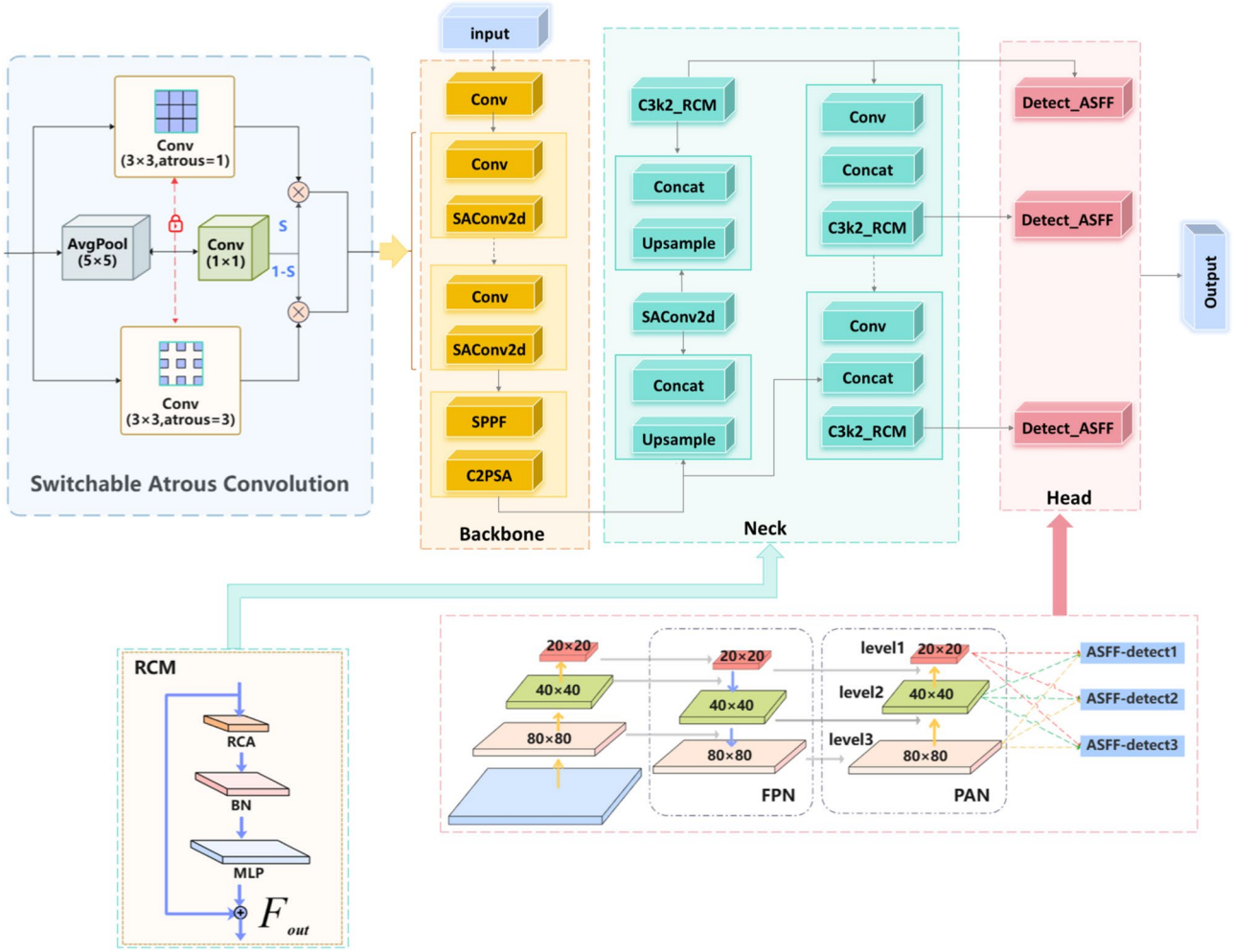


Fig. 1. YOLOv11-SRA Model Architecture Diagram.

layers with SAConv (switchable dilated convolution). A gated network dynamically selects convolution kernels with three dilation rates (1, 3, and 5) to improve the ability of the model to capture local features for multiscale safety helmet targets. The neck network reconstructs the fusion path of the feature pyramid by introducing a bidirectional long short-term memory (Bi-LSTM) cross-layer feature interaction module within the PAN-FPN architecture. Through gated recurrent units, it explores the multiscale contextual associations between adjacent layers and employs deformable convolution to replace the traditional upsampling interpolation process, thereby alleviating the feature misalignment caused by blurring in underground images. The detection head integrates the ASFF (adaptive spatial feature fusion) mechanism, which dynamically learns the weight distribution of underground safety helmet targets across different feature layers using a three-level spatial attention network. Coupled with a residual compensation strategy, this approach enhances the edge feature responses produced for small targets, thereby improving the localization accuracy achieved in dense scenes. Experiments demonstrate that this network model achieves closed-loop optimization from feature extraction and fusion to detection through geometric feature calibration in the RCM, dynamic receptive field adjustment in the SAConv layer, and cross-scale feature collaboration in the ASFF mechanism. This effectively addresses the false negatives and false positives caused by scale changes, occlusion interference, and low contrast levels in underground safety helmet detection scenarios.

SAConv: switchable atrous Convolution module

In object detection tasks, objects with varying scales have significantly different receptive fields requirements. The traditional convolution operations often struggle to dynamically adapt to these multiscale variations. To address the challenge of effectively capturing the features of multiscale safety helmets in underground environments, the Switchable Atrous Convolution(SAConv) module introduces a dynamic atrous rate selection mechanism alongside a spatially guided attention-based feature fusion strategy. This innovative approach involves the construction of an adaptive feature extraction structure that efficiently captures multiscale features and adapts to variations in the sizes of safety helmets.

The design philosophy of the SAConv module is grounded in a dynamic feature fusion strategy, centering on the intelligent selection of the optimal dilation rate configuration via a learnable switching mechanism. The

architecture of the module comprises three essential components: multibranch dilated convolution units, an adaptive weight generation network, and a global context enhancement module. As illustrated in Fig. 1, the module initially extracts features from the input feature map using a standard convolution with a dilation rate of 1, followed by a dilated convolution with a dilation rate of r. The dilated convolution expands the effective receptive field of a k × k convolution kernel to

$$\mathrm{k_e} = k + (\mathrm{k} - 1) * (\mathrm{r} - 1) \tag{1}$$

by incorporating r-1 zero padding, thereby enabling the capture of a broader range of contextual information without increasing the number of required parameters.

The middle SAC component incorporates two sets of convolutional operations, where the dilation rate r serves as a hyperparameter and Δ w denotes the trainable weight difference. The switching function $\mathrm{S}(\bullet)$ is implemented via a 5×5 average pooling layer and a 1×1 convolutional layer, generating spatially-correlated weight masks that control the process of selecting dilated convolution results at different positions.

The global context module introduces global context branches prior to and following the SAConv component. The SAConv structure is shown in Fig. 2. It compresses the input features using global average pooling and subsequently enhances the interchannel dependencies through a 1×1 convolution layer. The resulting features are then superimposed onto the primary feature stream, which operates analogously to a lightweight attention mechanism.

In the backbone network, all 3×3 convolutional layers can be substituted with SAC modules. Additionally, deformable convolutions are used to optimize the offset prediction. A weight-locking mechanism guarantees the production of consistent initial weights across various dilated convolutions, with only Δ w being fine-tuned to address any discrepancies. This methodology improves the stability of training while circumventing the computational burden of retraining the model from the ground up.

RCM rectangular self-calibration module

The original YOLOv11 neck network may exhibit deficiencies in terms of integrating feature information during feature fusion, particularly with respect to retaining and utilizing the features of small objects. To address this issue, the RCM (rectangular self-calibration module) is incorporated into the neck network for optimization purposes. This module is integrated after the backbone network, and employs horizontal and vertical pooling to extract the axial global context. It is further enhanced with large-kernel striped convolutions to calibrate the attention regions, enabling the network to concentrate on the critical areas of the head.

As shown in Fig. 3, RCM comprises three components: rectangular self-calibration attention (RCA), batch normalization (BN), and multilayer perceptron (MLP). It captures the global context through axial pooling and dynamically adjusts the attention region via a shape self-calibration function. For the input feature map, the module initially extracts axial global features through horizontal and vertical pooling operations, resulting in two axial vectors. Following the addition of these axial vectors, a large-kernel striped convolution is employed to calibrate the shapes of the attention regions, aligning them more effectively with the geometric distribution of the foreground target, thus enhancing the boundary localization capabilities of the model. The associated mathematical expression is as follows:

$$\mathrm{F_{out}} = X + MLP\left(\mathrm{BN}\left(\mathrm{RCA}\left(\mathrm{X}\right)\right)\right) \tag{2}$$

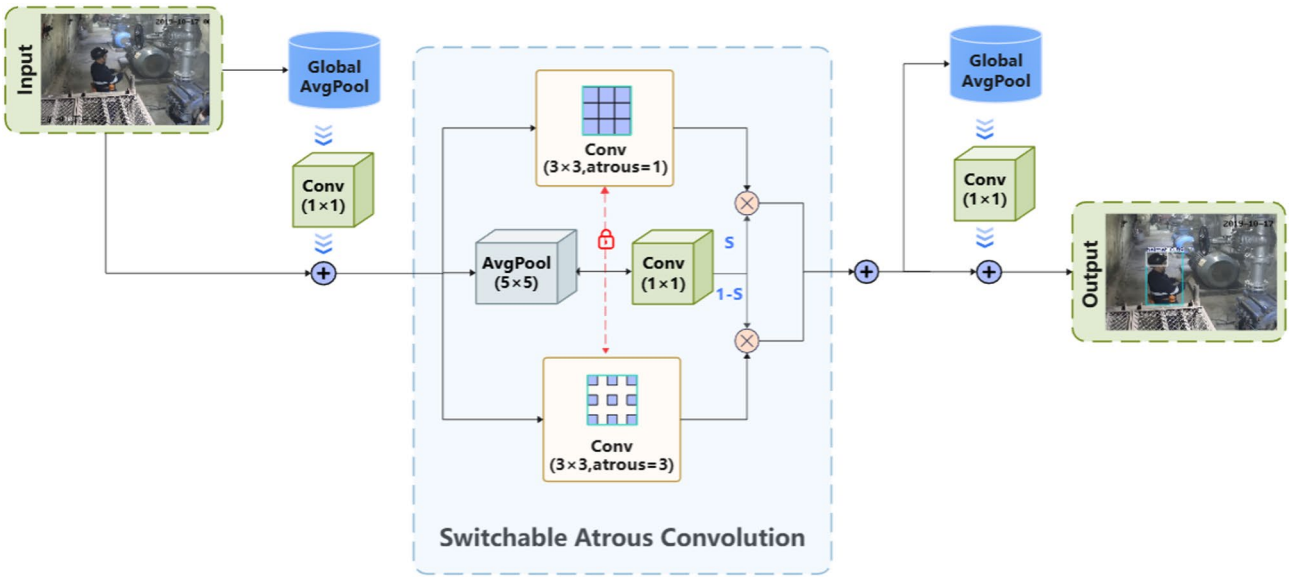


Fig. 2. Switchable Attenuated Convolution (SAConv).

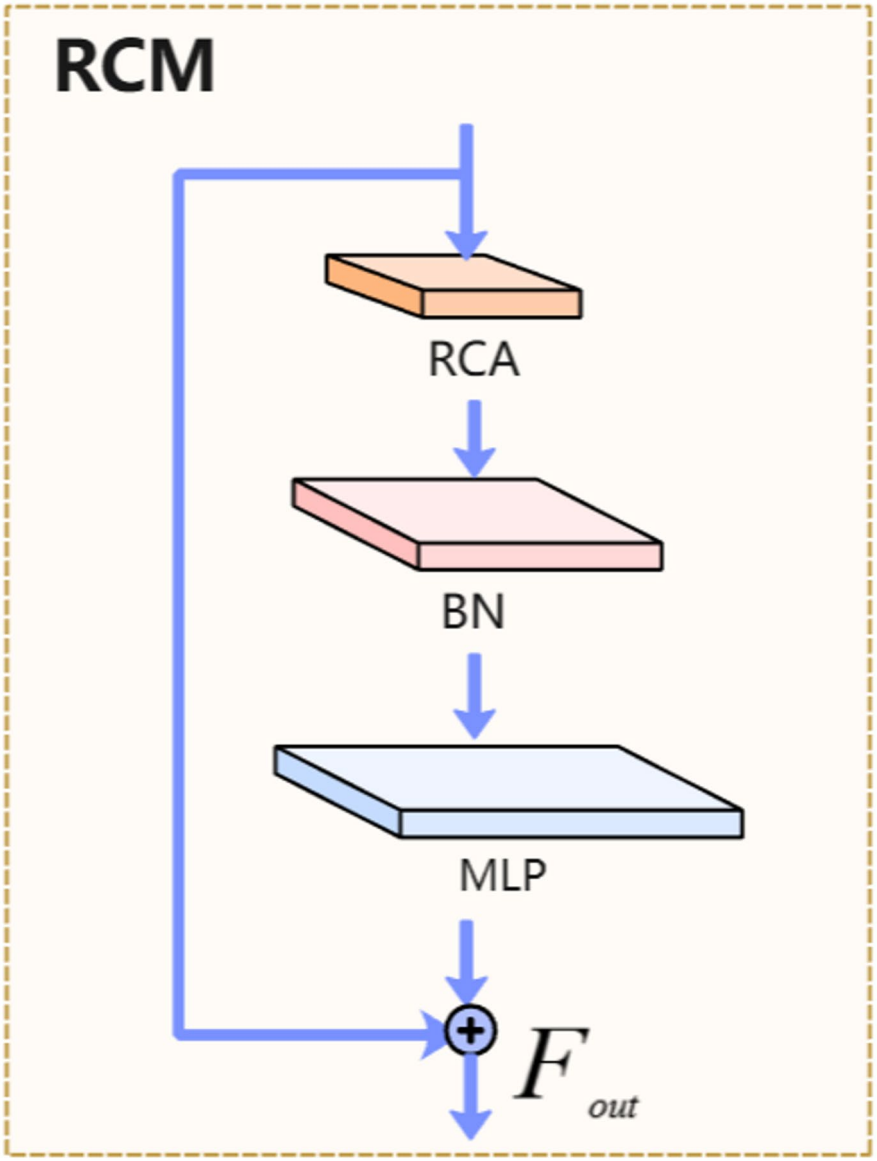


Fig. 3. RCM Module Structure.

The output features of the rectangular self-calibration module (RCM) are derived from the input features, which typically consist of feature maps obtained from the preceding layer of the neural network. These feature maps encapsulate a variety of semantic and spatial information pertaining to the given image. The multilayer perceptron (MLP) subsequently performs additional nonlinear transformations on the features following the batch normalization step. This process is aimed at learning more complex feature representations, thereby enhancing the capacity of the model to extract and articulate the target features.

ASFF: multiscale feature fusion module

In single-stage object detection tasks, inconsistencies encountered during multiscale feature fusion significantly hinder the detection performance of the utilized model. Heuristic-guided feature selection is typically employed, wherein larger objects are linked with upper-level feature maps, whereas smaller objects correspond to lower-level feature maps. When an object is classified as a positive sample within a specific feature map layer, the associated regions in other layers are designated as background layers. This results in feature conflicts that disrupt the gradient calculations and diminish the effectiveness of the feature pyramid. The adaptive spatial feature fusion (ASFF) technique employs adaptive weight learning and feature alignment mechanisms, allowing the network to spatially filter features directly from other layers directly. This approach retains only the pertinent information for combination purposes, effectively addressing the issue of blurred edges concerning in safety helmets within low-resolution underground images.

The core innovation of adaptive scale feature fusion (ASFF) mechanism lies in its ability to dynamically filter and fuse feature information across different scales by learning spatial adaptive weights. This approach effectively

suppresses the inconsistencies between features and enhances their scale invariance. Initially, the module adjusts the resolutions and channel counts of the feature maps at various levels through upsampling and downsampling operations, thereby achieving spatial alignment. During the upsampling process, a 1×1 convolution layer is applied first to reduce the channel count, and this is followed by bilinear interpolation to increase the resolution. Conversely, for downsampling, a 3×3 convolution layer with a stride of 2 is employed to simultaneously adjust both the number of channels and the resolution. For 1/4-scale downsampling, a maximum pooling layer is incorporated prior to the stride convolution to ensure more precise scale matching. Subsequently, a 1×1 convolution is utilized to generate spatial weight maps, which dynamically control the contribution ratios of the features at different scales, as illustrated in Formula 3:

$$y_{ij}^{l} = \alpha_{ij}^{l} \bullet x_{ij}^{1\to l} + \beta_{ij}^{l} \bullet x_{ij}^{2\to l} + \gamma_{ij}^{l} \bullet x_{ij}^{3\to l} \tag{3}$$

Where α_{ij}^{l}, β_{ij}^{l} and γ_{ij}^{l} are the spatially adaptive importance weights learned by the network, $\alpha_{ij}^{l} + \beta_{ij}^{l} + \gamma_{ij}^{l} = 1$ and $\alpha_{ij}^{l} ? \beta_{ij}^{l} ? \gamma_{ij}^{l} \in [0,1]$. Weight scalar maps α^{l}, β^{l} and γ^{l} are calculated from $x^{1\to l}$, $x^{2\to l}$ and $x^{3\to l}$, respectively, and learned through standard backpropagation. This pixel-wise weight learning mechanism enables the network to autonomously select the optimal feature combination at each spatial location, effectively suppressing feature conflicts and reinforcing useful information.

As illustrated in Fig. 4, deploying the ASFF module across three detection levels allows the network to deliver optimal feature representations for targets with varying scales, thereby significantly enhancing the performance achieved in multiscale target detection tasks. This approach effectively integrates semantic information and spatial details at multiple levels, facilitating a more accurate target localization process and ensuring stable detection performance.

Synergistic mechanism of the Three-Module architecture

The YOLOv11-SRA model addresses the multifaceted challenges of underground helmet detection through a meticulously designed collaborative architecture that integrates SAConv, RCM, and ASFF modules into a complementary optimized pipeline. Underground mining environments present four core technical challenges that traditional detection methods struggle to address concurrently: multi-scale feature conflicts arising from varying helmet dimensions; geometric distortions caused by partial occlusions from mining equipment and support structures; cross-scale feature misalignment when fusing information from different network layers; and low-contrast boundary localization under adverse lighting conditions, including dust, fog, and shadows. Our architecture does not treat these challenges as isolated problems requiring separate solutions; rather, it recognizes their fundamental interconnectedness and addresses them through a coordinated feature refinement process, wherein each module's output serves as enhanced input for subsequent processing stages.

The SAConv module initiates this optimization chain by dynamically adapting to scale variations through its switchable dilation rate mechanism. Its learned switching function generates spatially variable dilation patterns, automatically selecting $r=1$ for small objects that require high-frequency boundary details and $r=5$ for large objects that benefit from long-range contextual dependencies. This scale-aware feature extraction yields representations that simultaneously capture local precision and global context. However, these multi-scale features exhibit differing spatial resolutions, which may cause alignment issues in subsequent processing. Building upon these rich features, the RCM module addresses the geometric distortions inherent in occluded helmet detection by employing Rectangular Self-Calibrating Attention. Unlike conventional square convolutional kernels that uniformly process spatial dimensions, the RCM decomposes attention into axial components through horizontal and vertical pooling. It then applies large-kernel striped convolutions to calibrate attention regions according to the elongated geometric patterns characteristic of partially visible helmets. This geometric prior enhances boundary localization capabilities even when only fragmented helmet parts are visible.

Architectural synergy is achieved through ASFF's adaptive spatial weighting mechanism, which resolves the fundamental incompatibility between SAConv's dynamic receptive field and RCM's fixed geometric calibration pattern. ASFF learns pixel-level importance weights, dynamically determining feature combination strategies for each spatial position. It assigns higher weights to large-expansion-rate features in low-contrast regions while prioritising fine-grained features in high-frequency edge areas requiring precise localisation. This adaptive

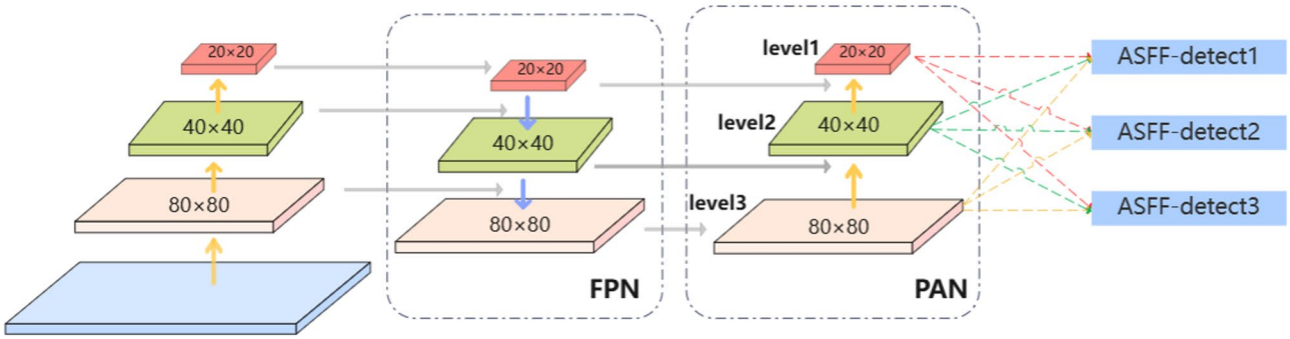


Fig. 4. Schematic Diagram of the Adaptive Spatial Feature Fusion Mechanism.

coordination eliminates feature conflicts and spatial misalignment that arise when directly stitching SAConv's multi-resolution outputs with RCM's geometrically calibrated features.

Experiments

Dataset and preprocessing

To validate the effectiveness of the proposed model, experiments and evaluations are conducted in this paper on the publicly available dataset CUMT-HelmeT dataset, as illustrated in Fig. 5.

Owing to the limited size of the dataset, this study employs data augmentation strategies to increase the size of the dataset. The original CUMT-HelmeT dataset, which comprises 2,482 images, is augmented through techniques such as cropping, translation, rotation, overlaying, and brightness adjustment. By combining these methods, the dataset is expanded to five times its original size, resulting in a total of 12,410 high-quality images. The dataset comprises six categories: no-helmet, helmet, car2, hat, *per*son, and people. The key "no-helmet" category contains the most samples (1,592 images), while the helmet category includes 1,006 images. This augmentation scheme not only effectively addresses the class imbalance issue that is prevalent in small-sample training cases but also significantly improves the adaptability of the model to real-world application scenarios by simulating complex underground lighting conditions and multiangle morphological variations in the target objects.

The augmented dataset is integrated with the original dataset, and subsequently partitioned into training, validation, and test sets at a 7:2:1 ratio through proportional partitioning. The training set facilitates parameter learning, allowing the model to identify the characteristic patterns of foreign objects within a comprehensive dataset. The validation set is utilized to dynamically adjust the model hyperparameters during training, thereby preventing overfitting or underfitting. The test set serves as a final evaluation of the model's performance of the model, ensuring the objectivity and accuracy of the evaluation results.

Experimental environment and parameter settings

To ensure the reliability and reproducibility of the experiments, this study utilized an NVIDIA RTX 4090D experimental platform, with key parameters configured appropriately. The software environment for model construction and training was based on Python 3.12.0, PyTorch 2.5.1, and CUDA 12.4. Detailed configurations of the experimental environment and training parameter settings are presented in Table 1.

Evaluation metrics

This experiment employs mean average precision (mAP), recall, and precision as evaluation metrics to comprehensively assess the detection performance of the model. The mean average precision (mAP) is a crucial metric for evaluating the performance of object detection models, as it considers the detection accuracy attained across different categories and provides an intuitive reflection of the overall detection capabilities of models. Recall measures the proportion of positive samples detected by the model out of all true-positive samples, serving as an important metric for evaluating the detection comprehensiveness of the model. A high recall rate

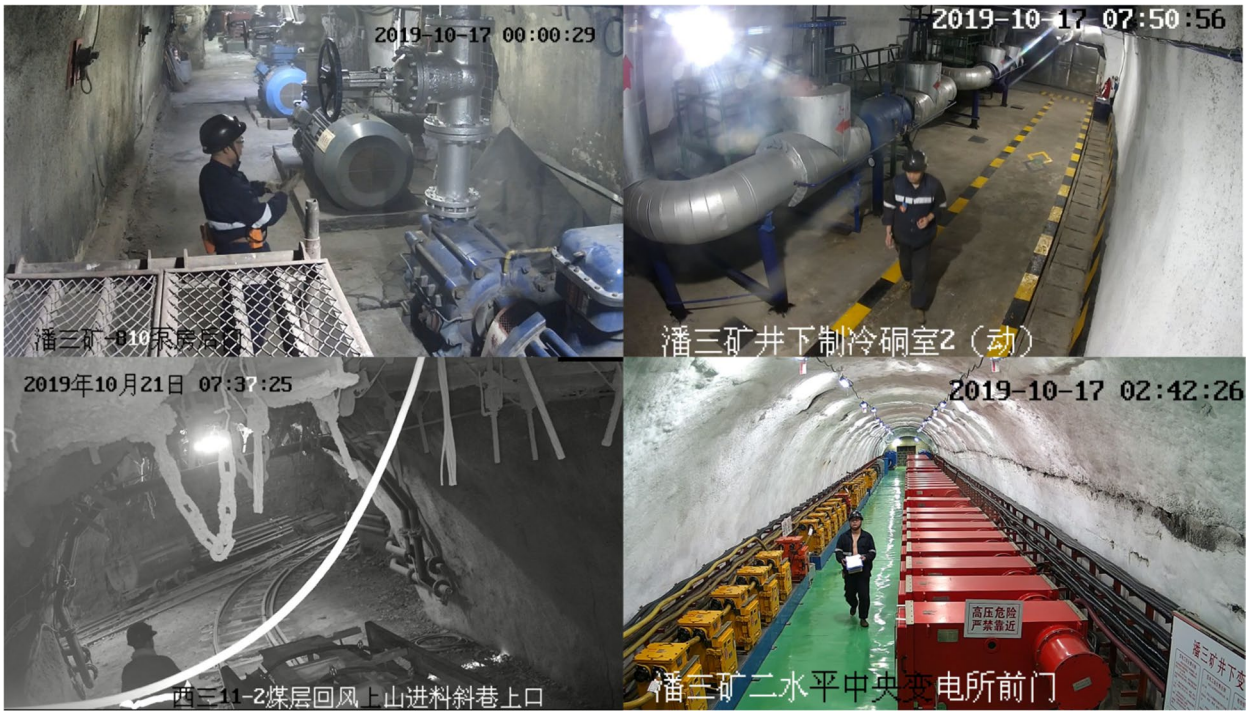


Fig. 5. Dataset Legend.

CPU	18 vCPU AMD EPYC 9754 128-Core Processor
GPU	RTX 4090D
RAM	24GB
Operating System	Ubuntu22.04
Programming Language	Python 3.12.0
Deep Learning Framework	PyTorch 2.5.1
GPU Acceleration Library	CUDA 12.4
Epochs	100
Batchs Size	32
Image Size Optimizer Learning_rate Momentum Weight_decay Box Loss Gain CLS Loss Gain DFL Loss Gain	640*640 SGD 0.01 0.973 0.0005 7.5 0.5 1.5

Table 1. Environment and Parameters.

indicates that the model can detect more target objects, thereby reducing the number of missed detections. In this experiment, we will adjust the detection threshold of the model to observe any recall changes and evaluate its performance at varying detection sensitivities. Precision measures the proportion of true-positive samples among the positive samples detected by the model and is a key metric for assessing the detection accuracy of the model. The mathematical definition of the given evaluation indicators is as follows:

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP+FP}} \tag{4}$$

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP+FN}} \tag{5}$$

$$mAP = \frac{\sum \mathrm{AP}}{\mathrm{N}} \tag{6}$$

In this study, we analyze the performance metrics of a classification model, focusing on true positives (TPs), false positives (FPs), and false negatives (FNs). True positives (TPs) are defined as the number of correctly predicted positive instances, while false positives (FPs) represent the number of instances that are incorrectly classified as positive. Conversely, false negatives (FNs) are positive instances that the model fails to identify. By evaluating these metrics, we can gain insights into the accuracy and effectiveness of the model in terms of making predictions.

Results

Evaluation YOLOv11-SRA

The training dynamics and performance metrics of the proposed YOLOv11-SRA model are illustrated in Fig. 6, which depicts the evolution of the loss function and evaluation metrics over 100 training epochs. A comprehensive analysis indicates that the YOLOv11-SRA model demonstrates robust learning capabilities and excellent convergence characteristics in its application to detecting safety helmets in coal mine shafts.

All loss components exhibited rapid and stable convergence during the training phase. The bounding box loss (train/box_loss) decreased significantly from an initial value of 3.5 to approximately 0.5 within the first 20 epochs. It subsequently stabilized around 0.5 throughout the remaining training iterations, showing minimal fluctuation. This indicates that the model efficiently learned to predict accurate bounding box coordinates for helmet detection. The classification loss (train/cls_loss) consistently decreased from 4.8 to below 0.5 during the early training stages, ultimately converging to approximately 0.4. The distribution focus loss (train/dfl_loss) exhibited a similar convergence pattern, steadily decreasing from 4.0 to around 0.9, reflecting the model's enhanced capability in predicting precise target locations.

The validation loss curve exhibits a consistent trend with the training loss. The validation bounding box loss (val/box_loss) stabilizes at approximately 1.25, the classification loss (val/cls_loss) stabilizes at 0.85, and the distribution focus loss (val/dfl_loss) stabilizes at 1.45. The relatively small gap between the training and validation losses indicates the YOLOv11-SRA model's strong generalization capability, which is essential for practical deployment in real underground mining environments characterized by significant variations in environmental conditions.

The evaluation metrics provide a comprehensive assessment of the model's detection capabilities. The precision (B) metric increases significantly from 0.4 to approximately 0.8 within the first 10 epochs, subsequently maintaining a stable range of 0.80 to 0.82 during the remainder of the training process. Similarly, the recall (B) metric shows a comparable improvement trajectory, reaching around 0.80 and stabilizing between 0.78 and 0.82, with minor fluctuations. The metrics/mAP50 (B) converged to approximately 0.82, indicating that the model achieved 82% mean average precision at an IoU threshold of 0.5. Additionally, the metrics/mAP50-95 (B) reached a value of approximately 0.58, reflecting the model's robust performance across a diverse range of IoU

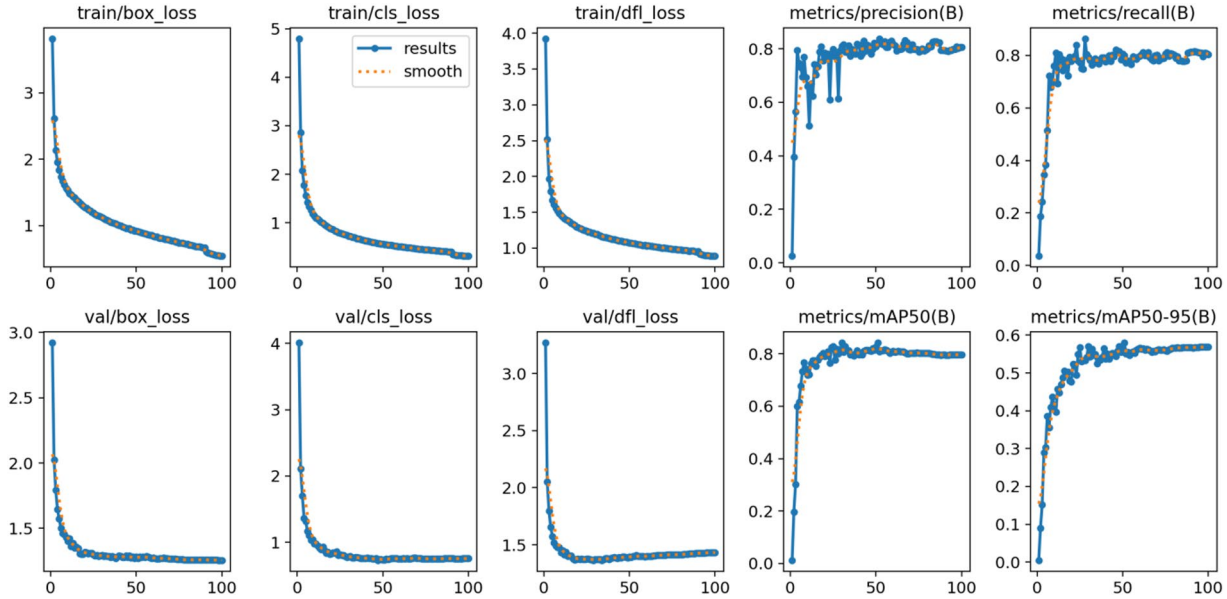


Fig. 6. Training and validation metrics for YOLOv11-SRA.

Model	Precision	Recall	mAP50
YOLOv9	0.768	0.652	0.774
YOLOv10	0.789	0.573	0.619
SSD	0.504	0.367	0.442
RetinaNet	0.646	0.486	0.531
Faster R-CNN	0.757	0.647	0.589
DETR	0.634	0.718	0.593
YOLOv11 + SAConv + C3k2_RCM + ASFF	0.817	0.799	0.842

Table 2. Comparison among the detection performances of different Models.

thresholds from 0.5 to 0.95. These metrics demonstrate the exceptional detection performance of the YOLOv11-SRA model, achieving an optimal trade-off between detection completeness and accuracy.

The training process demonstrated remarkable stability. Throughout all loss plots, the orange smoothing curve closely followed the original blue curve, showing no signs of instability or fluctuations during the optimization process. After the 50th epoch, both training and validation metrics converged to optimal values without divergence, indicating that the model successfully learned generalizable features capable of representing hardhat detection in diverse underground coal mine environments.

In summary, the YOLOv11-SRA model achieves a necessary balance between accuracy, precision, and recall, which is essential for real-time safety monitoring applications in coal mine shafts. The combination of high precision and high recall ensures reliable detection of safety helmets, thereby minimizing both false positives and false negatives. The stable mean Average Precision (mAP) score demonstrates the model's robustness under stringent evaluation criteria and its rapid adaptability, confirming its suitability for deployment in diverse mining environments characterized by poor lighting, dusty conditions, and complex backgrounds.

Comparative experiments

To comprehensively evaluate the detection performance of the improved model, mainstream object detection models, including YOLOv9, YOLOv10, RetinaNet, and the SSD, are selected for comparison. The comparison results are shown in Table 2.

Both the mainstream models and our proposed improved model demonstrate comparable performance in terms of detecting safety helmets underground. However, our improved model achieves significant enhancements across multiple key metrics, indicating its superior robustness and adaptability in practical applications. Although the mainstream models such as YOLOv9, YOLOv10, the SSD, RetinaNet, and Faster R-CNN exhibit commendable accuracy and recall rates, their generalization capabilities in complex environments require further enhancements. In contrast, our improved model, which incorporates the ASFF (adaptive spatial feature fusion) module, the SAConv (switchable dilated convolution) module, and the RCM (rectangular self-calibration module), not only significantly improves the resulting detection accuracy but also excels with respect to robustness and adaptability. This advancement is crucial for accurately detecting safety helmets in

Model	Parameters(M)	GFLOPs	FPS	Inference Time(ms)
YOLOv9	25.3	102.1	68.4	14.6
YOLOv10	15.4	59.1	87.3	11.5
SSD	25.1	241.6	56.8	17.6
RetinaNet	37.2	147.3	44.2	22.6
Faster R-CNN	141.8	375.2	30.5	32.8
DETR	42.7	88.4	34.3	29.2
YOLOv11 + SAConv + C3k2_RCM + ASFF	12.8	48.3	96.7	10.3

Table 3. Comparison of inference performance among different Models.

Baseline	SAConv	C3k2_RCM	ASFF	mAP50
√				0.592
√	√			0.783
√		√		0.553
√			√	0.624
√	√	√		0.442
√	√		√	0.651
√		√	√	0.757
√	√	√	√	0.842

Table 4. Comparison among the results of ablation Experiments.

underground environments, which are characterized by complexity and variability, with numerous lighting conditions and background interference types, thus imposing greater demands on the detection performance of the utilized model. Our improved model also excels in terms of the mAP50 metric, further validating its effectiveness in practical applications. Comparative experiments indicate that while the mainstream models perform well in terms of specific aspects, our improved model has a more comprehensive advantage in terms of overall performance, offering broader application prospects and higher practical value in the task of detecting safety helmets in complex underground environments.

To address the real-time deployment requirements in underground environments, we conducted comprehensive inference performance evaluations on the same hardware platform.

As shown in Table 3, our YOLOv11-SRA model achieves outstanding performance across all efficiency metrics, with a parameter count of merely 12.8 million—approximately 17% fewer than YOLOv10 and over 69% fewer than the representative two-stage detector, Faster R-CNN. Concurrently, it exhibits the lowest computational complexity at 48.3 GFLOPs, representing reductions of approximately 18% and 87% compared to YOLOv10 and Faster R-CNN, respectively. These reductions fully demonstrate the effectiveness of the integrated lightweight modules in maintaining expressive capability while reducing resource demands.Regarding inference speed, the proposed model achieves a peak of 96.7 FPS with a latency of merely 10.3 ms. This represents a 10.7% improvement in FPS over YOLOv10 and nearly triples the performance of Faster R-CNN, with corresponding latency reductions of 10.4% and 68.6%. This outstanding real-time performance stems from the synergistic effect of efficient convolutional design and attention mechanisms, optimizing the feature processing pipeline without compromising robustness.Compared to existing mainstream approaches, the proposed YOLOv11-SRA model achieves a superior trade-off between model size, computational overhead, and inference speed. This advantageous characteristic renders it particularly suitable for resource-constrained underground deployment scenarios, delivering enhanced reliability in practical operations where low latency demands are paramount.

Ablation experiments

To comprehensively validate the effectiveness of each enhanced module with regard to improving the performance of the model, ablation experiments are conducted on the proposed improved model utilizing the preprocessed CUMT-HelmeT dataset. These experiments are grounded in the YOLOv11 model, with enhancements systematically integrated through the SAConv (switchable hollow convolution) module, the RCM (rectangular self-calibration module), and the ASFF (multiscale feature fusion) module. This approach allows for the quantification of the contribution of each module to the detection accuracy and efficiency of the model. Table 4 presents the detailed results of the ablation experiments, with the baseline established as the YOLOv11 model.

The results of the ablation experiment are shown in Table 4. The results of the ablation experiments further substantiate the significant improvement in model performance that is attributed to each improved module. Utilizing the preprocessed CUMT-HelmeT dataset, we conduct comprehensive ablation experiments on the proposed enhanced model to quantify the contribution of each module to its detection accuracy and efficiency. Findings indicate that when deploying SAConv, ASFF, and ASFF modules independently, the model's mean average precision (mAP) metric consistently improves. However, when RCM was deployed alone or combined

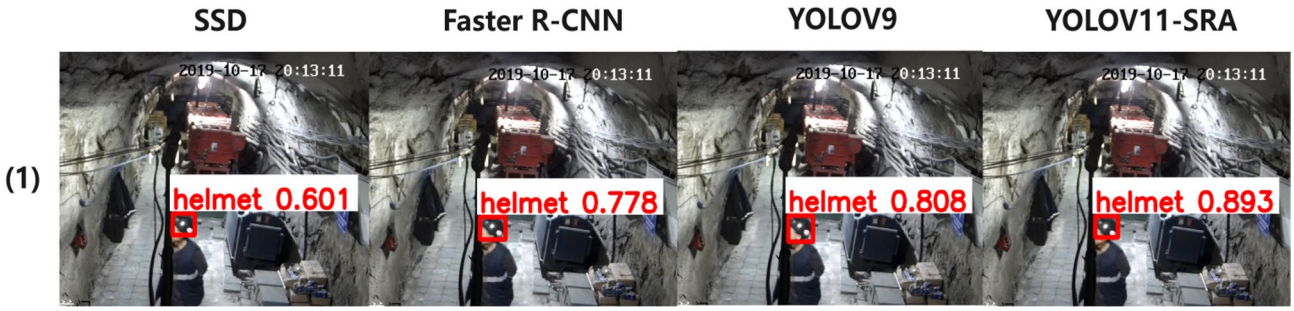


Fig. 7. Comparison of Model Detection Results in Severely Occluded Scenarios.

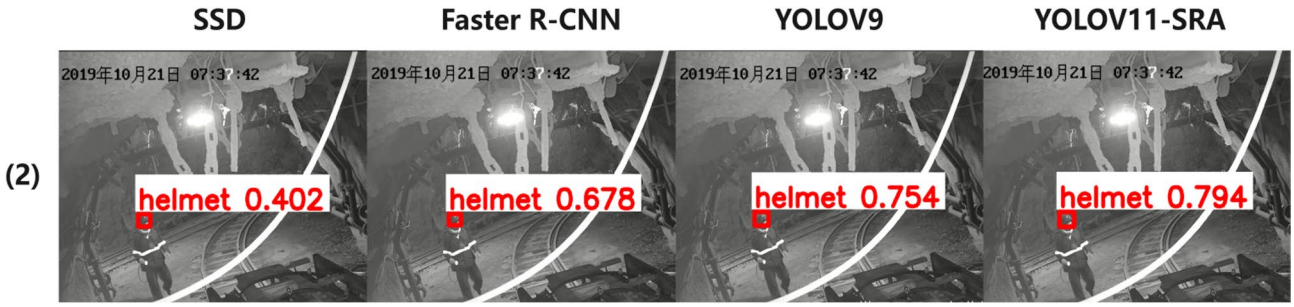


Fig. 8. Comparison of model detection results in extremely low-light scenarios.

with SAConv, an anomalous performance decline occurred. This phenomenon stems from the inherent design characteristics of RCM's rectangular self-calibration mechanism, which refines foreground regions through axial pooling and large-kernel striped convolutions. While this geometric calibration proves effective for isolating helmet boundaries, it requires rich multi-scale contextual information to function optimally. Without the contextual support provided by SAConv's adaptive scaling factor, the RCM module tends to over-calibrate attention regions, inadvertently suppressing valuable background cues essential for distinguishing helmet boundaries in cluttered underground environments. The situation worsens when SAConv and RCM operate without ASFF, creating a fundamental feature bottleneck. SAConv dynamically expands the receptive field and generates feature maps with varying spatial resolutions, while RCM applies geometric calibration using fixed rectangular kernels. This architectural mismatch introduces spatial misalignment in feature representations, disrupting gradient flow during backpropagation and ultimately hindering effective learning.

Synergistic effects emerge when all three modules are integrated. The ASFF module acts as a bridge, coordinating multi-scale features processed by SAConv with geometrically calibrated outputs from RCM through pixel-wise weight learning capabilities. This suppresses conflicting information and resolves incompatibilities. This achieves an overall improvement of approximately 40%, with the substantial performance gain primarily attributed to synergistic interactions between modules: the SAConv module enhances the model's ability to capture multi-scale features through switchable dilated convolutions; the C3k2_RCM module improves local feature representation via a rectangular auto-calibration mechanism; while the ASFF module optimises multi-scale feature fusion through adaptive spatial feature integration. Ablation studies demonstrate that integrating these modules not only elevates detection accuracy but also substantially enhances computational efficiency, thereby enabling superior real-time performance and robustness in practical applications.

Comparison among the experimental results

To visually demonstrate the advantages of the proposed method in complex underground environments, comprehensive visual comparison results are presented in Figs. 7, 8 and 9, and 10. These figures illustrate the detection performance under four typical challenging conditions commonly encountered in coal mining scenarios: severe occlusion, extremely low illumination, small target sizes, and strong reflections.

In the severe occlusion scenario, approximately 40–50% of the helmet area is obscured by equipment. As shown in Fig. 7, SSD achieves a confidence score of only 0.601, while the Faster R-CNN demonstrates moderate performance but remains insufficiently reliable for safety-critical applications. YOLOv9 shows improvement but retains some uncertainty. In contrast, our YOLOv11-SRA achieves the highest confidence score of 0.893, representing an 11.8% improvement over SSD and a 10.5% improvement over YOLOv9. This superior performance is attributed to RCM module, which decomposes spatial attention into horizontal and vertical components to effectively capture the elongated geometric patterns of partially visible helmets. The large-kernel striped convolutions within RCM align attention regions with actual foreground distributions, enabling robust detection even when only fragmented helmet areas are visible.

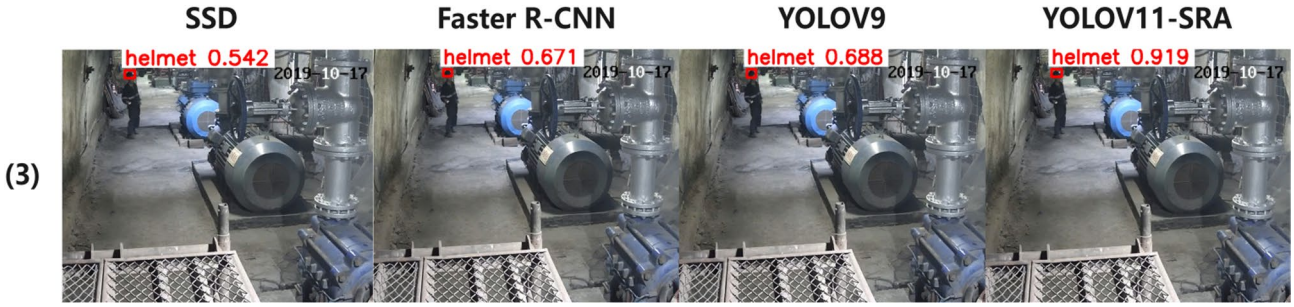


Fig. 9. Comparison of small object detection results.

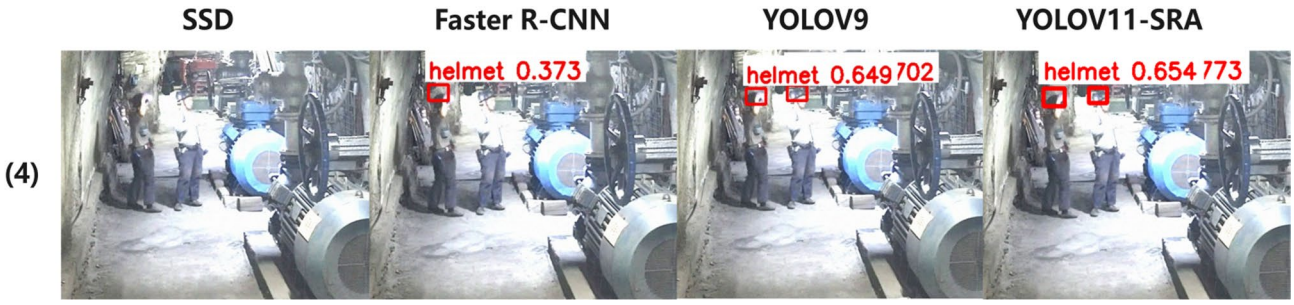


Fig. 10. Comparison of Strongly Reflective Scene Detection Results.

The results in Fig. 8 demonstrate that under extremely low light conditions with severe shading, the SSD performance sharply declined to 0.678. Faster R-CNN demonstrates higher robustness but remains insufficient for reliable deployment; YOLOv9 achieves a more reasonable detection rate of 0.754. Our YOLOv11-SRA model achieves the highest confidence score of 0.794, representing a 5.3% improvement over YOLOv9. This enhancement stems from the global context enhancement branch of SAConv, which strengthens weak edge responses in low-contrast regions through global average pooling and convolutions that augment inter-channel dependencies, compensating for the loss of local details in dark areas.

According to the results in Fig. 9, all three comparison models (SSD, Faster R-CNN, and YOLOv9) performed poorly in small object detection scenarios. YOLOv11-SRA demonstrated significantly higher confidence at 0.919, improving by 69.6% and 33.6% over SSD and YOLOv9 respectively. It effectively preserved critical boundary details prone to loss during feature pyramid fusion. The residual compensation strategy further enhances edge feature response, enabling precise localization of small helmet targets even in dense scenes.

Figure 10 clearly demonstrates that in highly reflective environments, direct illumination from mining lamps creates specular highlights on helmet surfaces. This phenomenon renders the SSD completely incapable of detecting the target due to the disruption of local feature consistency caused by intense reflections. Faster R-CNN detected only partial targets, while YOLOv9 achieved similar moderate performance—all struggling with local brightness abrupt changes. YOLOv11-SRA demonstrated the most stable detection results, maintaining a performance of 0.655. This stability stems from SAConv's adaptive expansion, which incorporates broader contextual information to mitigate local intensity interference.

By comparing detection results across the four challenging scenarios, the effectiveness of the collaborative design of SAConv, RCM, and ASFF modules is validated. These visualizations provide empirical evidence that the proposed structural innovations translate theoretical improvements into practical detection performance gains, demonstrating that YOLOv11-SRA is a robust solution suitable for real-world underground safety monitoring applications.

Conclusion

Underground safety helmet detection plays a critical role in ensuring mining safety. In this study, an efficient and accurate underground safety helmet detection system is constructed by introducing an adaptive spatial feature fusion module to optimize multiscale detection heads, adopting switchable hole convolutions to enhance the feature expression capabilities of the backbone network, and designing a rectangular self-calibration module to improve the structure of the neck network. On the preprocessed CUMT-Helmer dataset, the synergistic optimization effects of each module were validated through ablation experiments: the SAConv module improved the mAP by 19.1%, the RCM further enhanced target localization capabilities of the model, and the ASFF module achieved optimal fusion of cross-scale features fusion. The final integrated model achieved an mAP50 of 84.2%, representing a 42.2% improvement over that of the baseline model. Comparison experiments demonstrated that the model outperforms mainstream models such as YOLOv9 and RetinaNet in terms of the accuracy (81.7%)

and recall rate (79.9%) metrics. The experimental results confirm the detection advantages of the improved strategy for addressing occluded targets in complex underground scenarios. Future research will focus on two aspects: constructing a multi-illumination, multiview underground scene dataset to enhance the generalizability of the model, and exploring a fusion paradigm consisting of lightweight attention mechanisms and dynamic convolutions to overcome the detection accuracy bottleneck while maintaining high real-time performance.

Data availability

The datasets during the current study are available in the repository https://github.com/ww3312/YOLOv11-SRA-Helmet-Detection.git.

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Author contributions

L.W. conceived the study, designed the YOLOv11-SRA model architecture, and implemented the SAConv module. X.W. conducted the experiments, performed data preprocessing and augmentation, and carried out comparative and ablation studies. X.S. developed the RCM and ASFF modules, optimized the neck and detection head networks, and analyzed the experimental results. A.W. contributed to the writing and revision of the manuscript, organized the references, and assisted in the validation of experimental data.

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Declarations

Competing interests

The authors declare no competing interests.

Author contributions statement

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Accession codes: https://github.com/ww3312/YOLOv11-SRA-Helmet-Detection.git.

Additional information

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